How UltraBreak builds one universal image

Optimization on a white-box surrogate



Image patch
224 x 224 px



Random transform: rotate up to 15 deg, scale 0.8 to 1.2, random position on a base image

Surrogate VLM

Qwen2-VL-7B in the paper, plus a text prompt

Loss

Semantic loss on token embeddings (instead of cross-entropy), plus a smoothness penalty (TV)

Gradient step on the patch pixels, repeated over many prompts and epochs

Transfer to unseen models (black-box)



Final image

- Open VLM A
- Open VLM B
- Newer models (Axis 2)

Paper reports about 71% average success on held-out open models